

Single-particle tracking PALM analysis of CAZ proteins in neuronal synapses

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Goal: Develop a reproducible computational pipeline for single-particle tracking PALM analysis of CAZ protein mobility and dynamics under different cellular conditions.

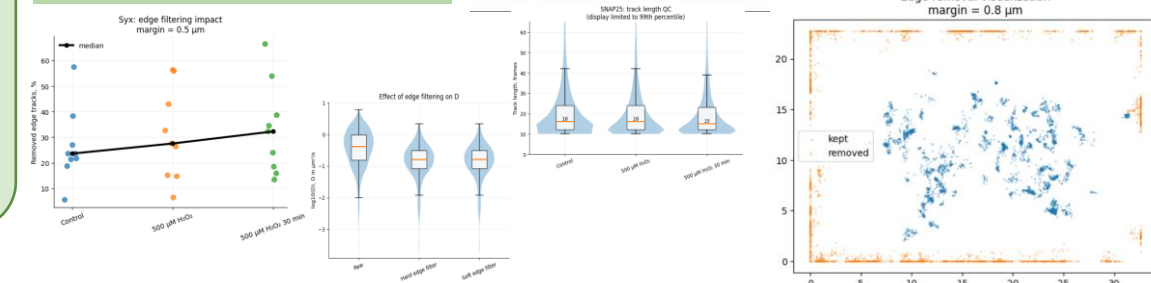
Data acquisition

- HEK293T / neurons
- CAZ protein expression (SNAP25b и Syntaxin)
- sptPALM imaging
- TrackMate export

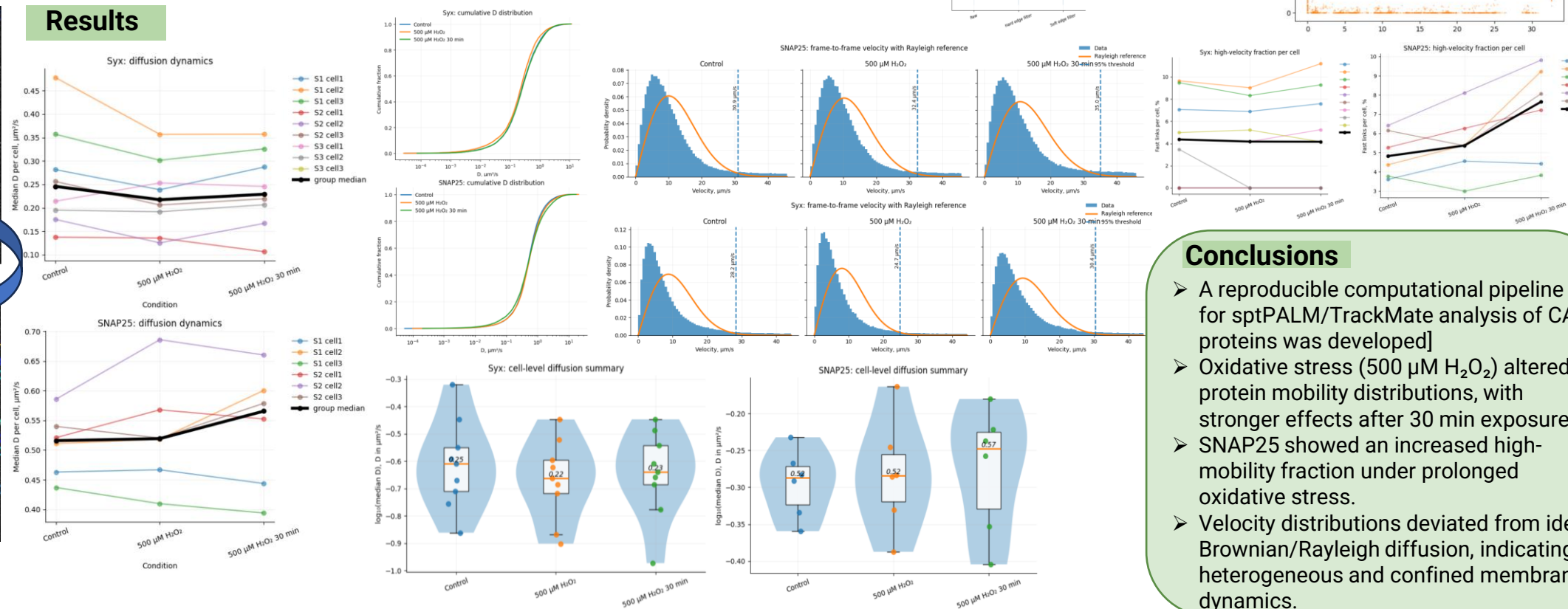
Processing pipeline

- **Particle detection & trajectory linking** (TrackMate CSV)
- **Edge filtering** (remove border artifacts)
- **Track QC** (length / displacement control)
- **MSD / D estimation** ($D = \text{slope} / 4$)
- **State classification** (slow / mobile fractions)
- **Statistics & visualization** (paired comparison across conditions)

QC and artifact control



Results



Conclusions

- A reproducible computational pipeline for sptPALM/TrackMate analysis of CAZ proteins was developed]
- Oxidative stress (500 μM H_2O_2) altered protein mobility distributions, with stronger effects after 30 min exposure.
- SNAP25 showed an increased high-mobility fraction under prolonged oxidative stress.
- Velocity distributions deviated from ideal Brownian/Rayleigh diffusion, indicating heterogeneous and confined membrane dynamics.